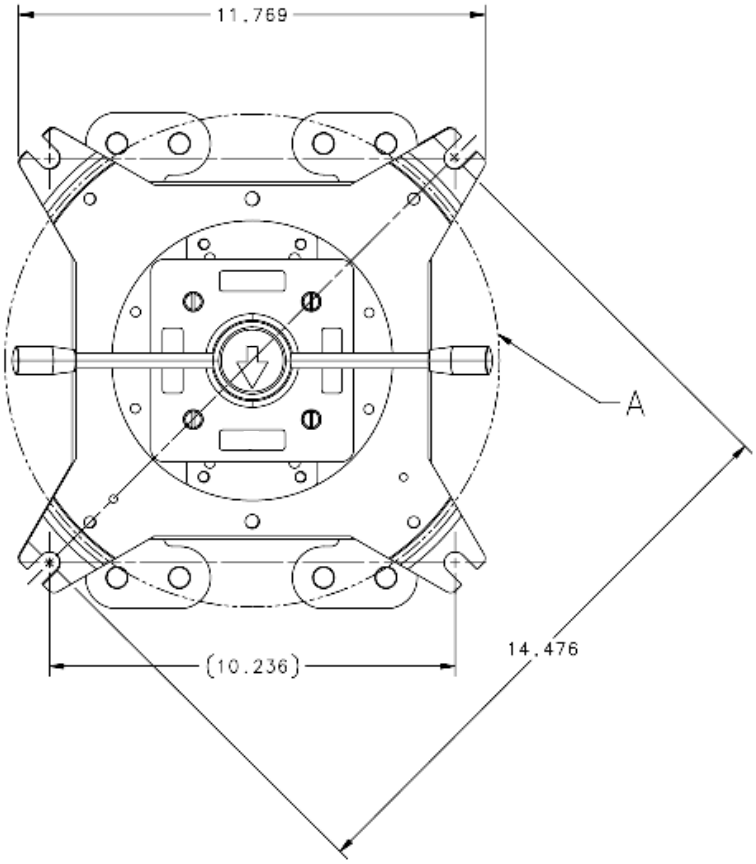
INCH-POUND
MIL-DTL-15291/14A
21 January 2022
SUPERSEDING
MIL-DTL-15291/14
7 May 2013

DETAIL SPECIFICATION SHEET

SWITCHES, ROTARY, SNAP ACTION CLASS 80SR FRONT AND REAR MOUNTED, SIDE CONNECTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the switch described herein shall consist of this specification sheet and MIL-DTL-15291.



NOTES
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is plus or minus 0.015

FIGURE 1 Class 80SR Switch Front View

AMSC N/A
FSC 5930
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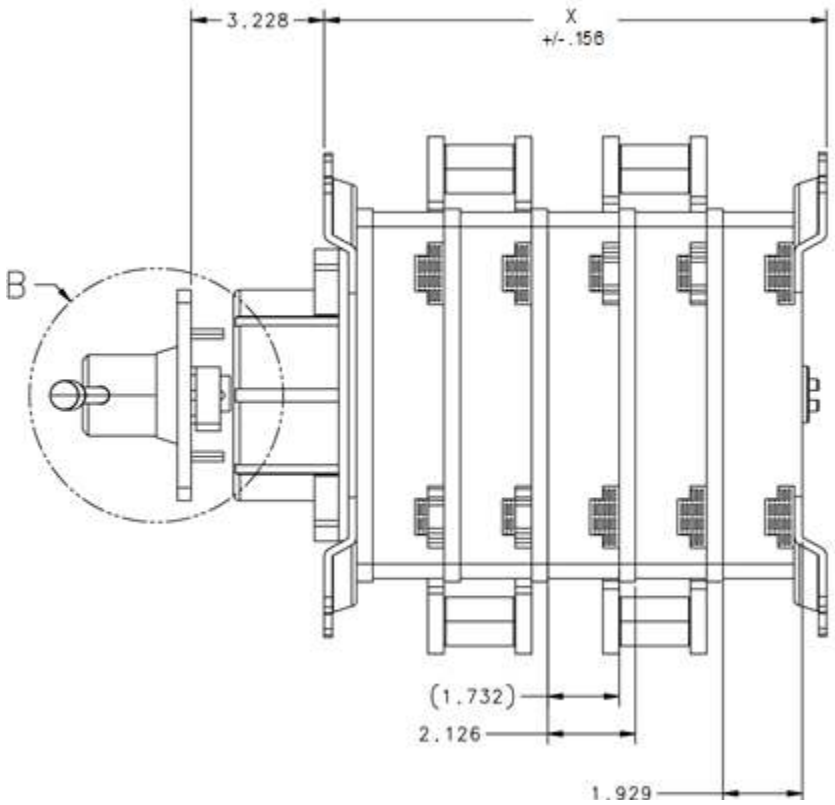


FIGURE 2 Class 80SR Switch Side View

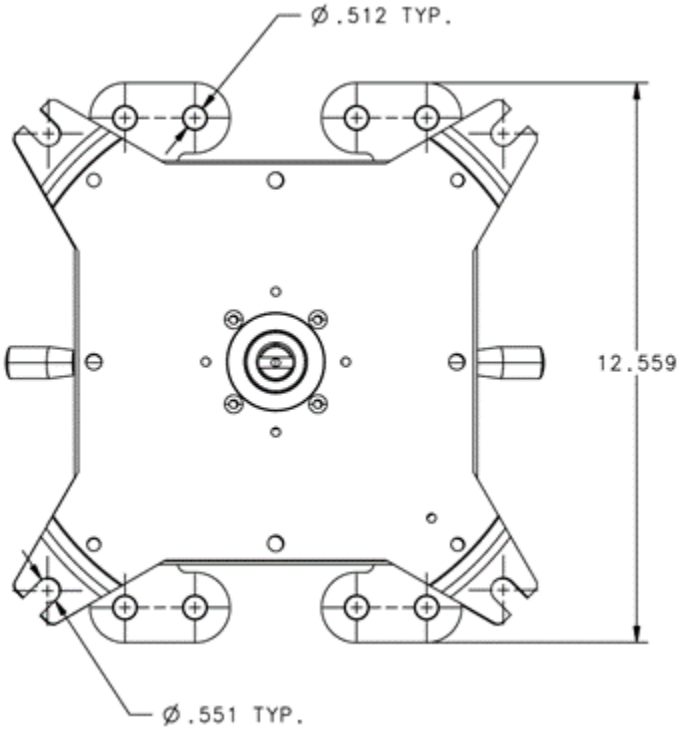


FIGURE 3 Class 80SR Switch Rear View

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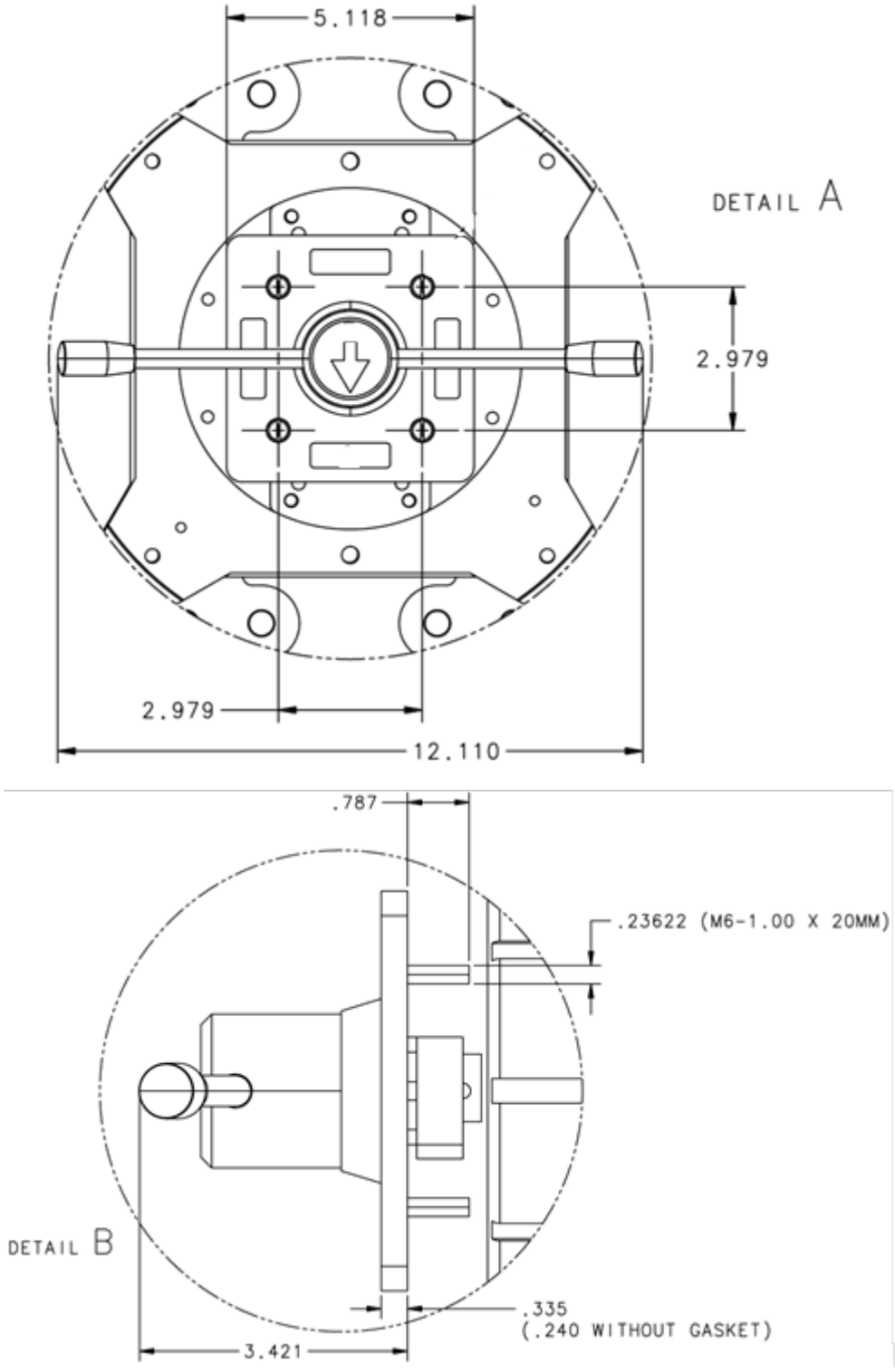


FIGURE 4 Class 80SR Standard Handle

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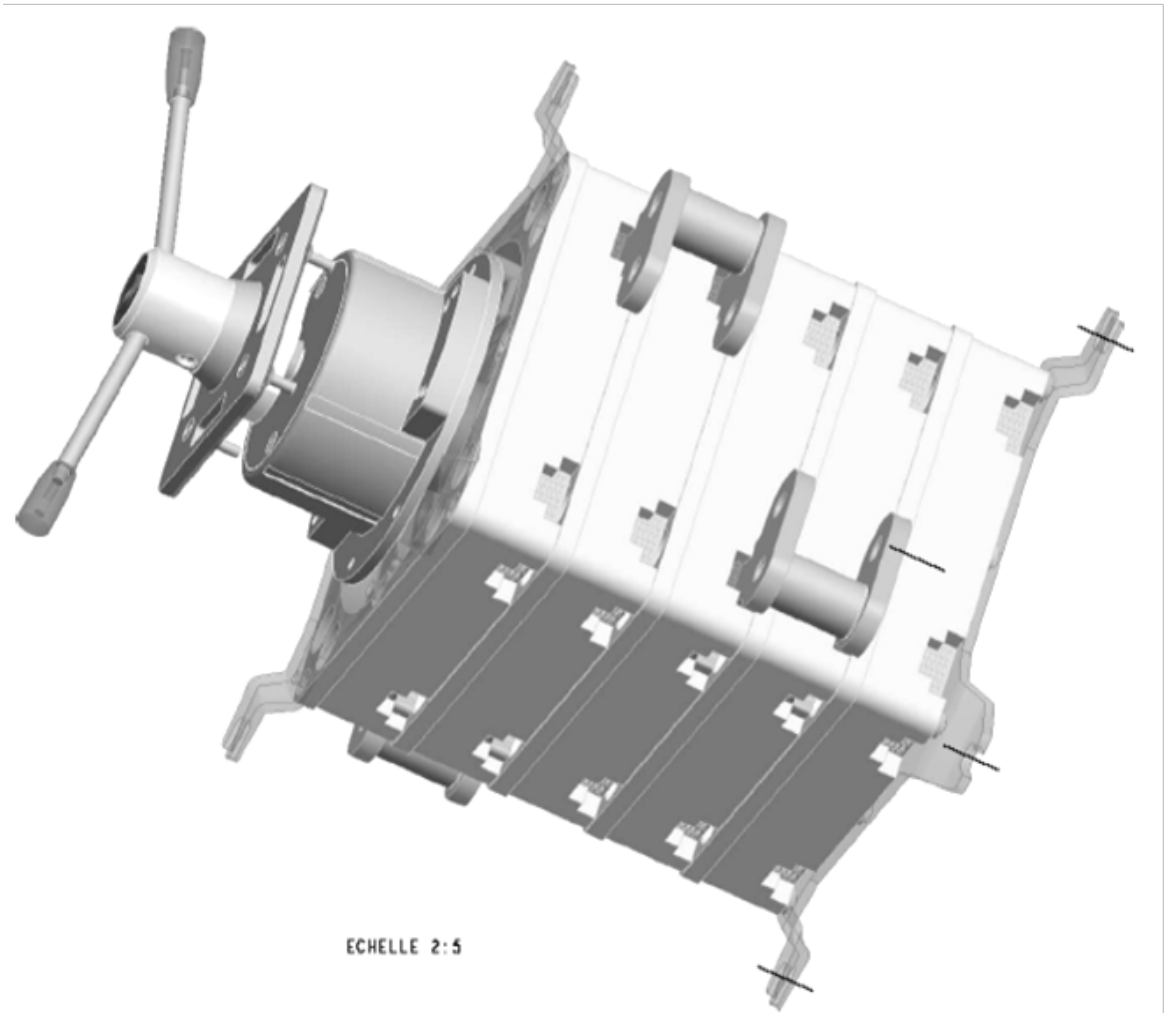
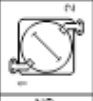
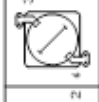
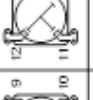
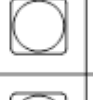
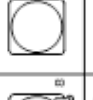
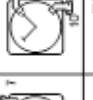


FIGURE 5 Class 80SR Three Dimensional View (Four Contacts)

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TABLE I. Type and switching characteristics.

M15291/14 DASH NO. TYPE DESIGNATION	DETAIL REF	TORQUE IN-LBS	DIM "X"	HANDLE OR SHAFT POSITION	CIRCUIT AND SPACER CONFIGURATION ROTOR POSITIONS, SPACER LOCATIONS, TERMINAL MARKING AND LOCATIONS									NOTES
					NO 1	NO 2	NO 3	NO 4	NO 5	NO 6	NO 7	NO 8	NO 9	
-001 80SR2A1	FIG 1	80 FT-LBS	10.079	OFF ON										
-002 80SR3A1	FIG 1	80 FT-LBS	12.205	OFF ON										
-003 80SR2A1A	FIG 1	80 FT-LBS	10.079	OFF ON										
-004 80SR3A1A	FIG 1	80 FT-LBS	12.205	OFF ON										
-005 80SR4F1	FIG 1	80 FT-LBS	14.331	OFF 1&2										
-006 80SR6F1	FIG 1	120 FT-LBS	18.583	OFF 1&2										
-007 80SR2S1	FIG 1	80 FT-LBS	14.331	OFF ON										
-008 80SR2E1	FIG 1	80 FT-LBS	14.331	OFF 1&2										
-009 80SR2S1A	FIG 1	80 FT-LBS	14.331	OFF ON										
-010 80SR2S1B	FIG 1	120 FT-LBS	18.583	OFF ON										
-011 80SR2S2	FIG 1	120 FT-LBS	22.835	OFF ON										
-012 80SR2E1	FIG 1	120 FT-LBS	18.583	OFF 2										

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REQUIREMENTS:

Applicable specification: MIL-DTL-15291

Dimensions and mounting: See figure 1

Switching characteristics: See table I

Angle of throw: 90 degrees

Switching action: Snap action, reciprocating

Transfer time: ≤ 30ms

Electrical and endurance ratings: See table II

Stop strength (applicable to switches with stops): 15 ft-pounds

Vibration: 50 Hz, MIL-STD-167-1

Shock (high impact): High impact, MIL-DTL-901

Contact resistance: 1.5 milliohms maximum

Dielectric withstanding voltage: 3,500 Vrms

Insulation resistance: 100 megohms minimum

Mounting bolts: (8) ½ inch diameter, length, and headstyle to suit application. Bolts not furnished.

Terminal marking: Terminal markings shown in table I locate terminals as viewed from the front of the switch (handle end). Markings shall be stamped on the switch body adjacent to the terminals.

TABLE II. Electrical and endurance ratings.

Tests	Current (amperes)	Voltage (volts)	Short Circuit Withstand (kA) (t ≤ 0.25sec)	Electrical Operations (number of operations)	Test rate (operations per minute)	De-energized operations (number of operations)
Alternating current (rms)[1]	800	500	100	4000	4	2000
Direct current[2]	800	400	100	1000	2	500

1 - Rating based upon one contact per phase.
2 - Rating based upon one contact per polarity.

GENERAL INFORMATION:

Switches not covered by Specification sheets: Switches which are fabricated from standard parts, as used in qualified switches, but which do not comply with switches detailed herein with respect to circuit characteristics, switching action, mounting arrangement, and handle details may be acquired under this specification from contractors having qualification approval under this specification.

Extended ratings: Switches detailed herein have been tested and found satisfactory at the extended ratings listed in table III. Reduced life expectancy must be anticipated for switches used at these increased voltage or current levels. Tests under the conditions of table III are not required for qualification acceptance and they are not repeated routinely as for maintenance of qualification. Supplemental evaluations and tests applicable to particular circuit requirements are recommended.

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TABLE III. Extended Ratings

Switching Character-istics	AC 60 or 400 Hz												
	125 volts				250 volts				500 volts				
	Resistive or Lamp load		Inductive load 0.75 p.f.		Resistive or Lamp load		Inductive load 0.75 p.f.		Resistive or Lamp load		Inductive load 0.75 p.f.		
	Amp.	Ops.	Amp.	Ops.	Amp.	Ops.	Amp.	Ops.	Amp.	Ops.	Amp.	Ops.	
	800	4000	800	4000	800	4000	800	4000	800	4000	800	4000	
	DC (All ratings based upon two poles (positive and negative) per circuit.)												
	24 volts				115 volts				440 volts				
	Resistive or Lamp load		Inductive load 0.75 p.f.		Resistive or Lamp load		Inductive load 0.75 p.f.		Resistive or Lamp load		Inductive load 0.75 p.f.		
	Amp.	Ops.	Amp.	Ops.	Amp.	Ops.	Amp.	Ops.	Amp.	Ops.	Amp.	Ops.	
	800	4000	800	4000	800	4000	800	4000	800	4000	800	4000	

Qualification inspection: Qualification inspection shall be performed at a laboratory acceptable to the Government on sample units produced with equipment and procedures normally used in production. Qualification shall consist of the examination and tests specified in table IV.

TABLE IV. Qualification Tests.

Inspection	Requirement per MIL-DTL-15291	Test Method per MIL-DTL-15291	Sample number		
			1	2	3
General Examination	3.5 and 3.7	4.6	1	1	1
Operation	3.6.1	4.7.1	2	2	2
Stop Strength	3.6.1.2	4.7.1.2		3	
Transfer Time	3.6.1.3	4.7.1.3	3	4	3
Salt Spray	3.6.2	4.7.2			4
Vibration Test	3.6.3	4.7.3			5
High-impact Shock	3.6.4	4.7.4			6
Dielectric Withstand	3.6.5	4.7.5	4	5	7
Contact Resistance	3.6.7	4.7.7	5	6	8
AC Endurance	3.6.9	4.7.9.1	6		
DC Endurance	3.6.9	4.7.9.2		7	
Temperature Rise	3.6.8	4.7.8	7		
Insulation Resistance	3.6.10	4.7.10	8	8	9
Operation	3.6.1	4.7.1	9	9	11
Transfer Time	3.6.1.3	4.7.1.3	10	10	12
General Examination	3.5	4.6	11	11	13

Sample:
1. Type 80SR3A1
2. Type 80SR2A1
3. Type 80SR2S2

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APPLICATION AND ACQUISITION GUIDE: PIN and type designation cross reference shall be as shown in table V

TABLE V. Application and acquisition guide.

M15291/14 Dash No.	Type Designation	For new or Existing design	For replacement	Circuit Configuration
-001	80SR2A1	X		A (on-off-on-off)
-002	80SR3A1	X		A (on-off-on-off)
-003	80SR2A1A	X		A (on-off-off-off)
-004	80SR3A1A	X		A (on-off-off-off)
-005	80SR4F1	X		F (off-on 1-on 1 & 2-on 2)
-006	80SR6F1	X		F (off-on 1-on 1 & 2-on 2)
-007	80SR2S1	X		S (on-off-off-off)
-008	80SR2B1	X		B (off-on 1-off-on 2)
-009	80SR2S1A	X		S (on-off-off-off)
-010	80SR2S1B	X		S (on-off-off-off)
-011	80SR2S2	X		S (on-off-on-off)
-012	80SR2E1	X		E (off-on 1-on 2-on 3)

Referenced Documents:
MIL-DTL-15291
MIL-DTL-901
MIL-STD-167-1

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Navy - SH
DLA - CC

Review Activities:
Navy - AS

Preparing activity:
DLA - CC

(Project 5930-2022-023)

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